Netaji Subhash Engineering College

PRESENTATION ON :-

1) Advantages of Database Management System over conventional file management System. [PCC-CS601.1] L1

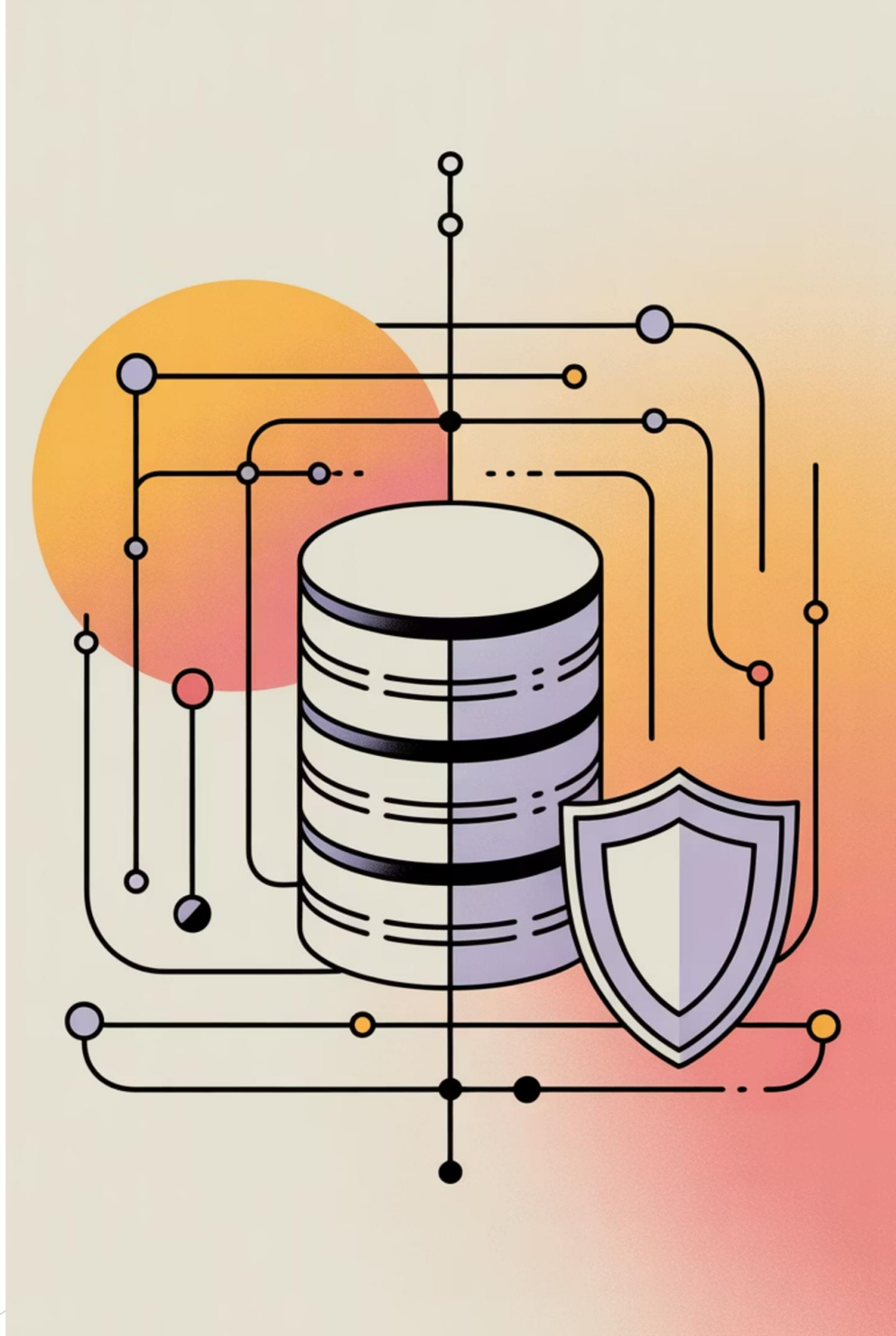
2) Three-level (tier) architecture of DBMS. [PCC-CS601.2] L2

Database Management Systems PCC-CS601(6577)

Name :- Rohan Karak
Section :- B
Semester :- 6th
University roll no. :- 10900123118
Class roll no. :- 117
Department :- CSE

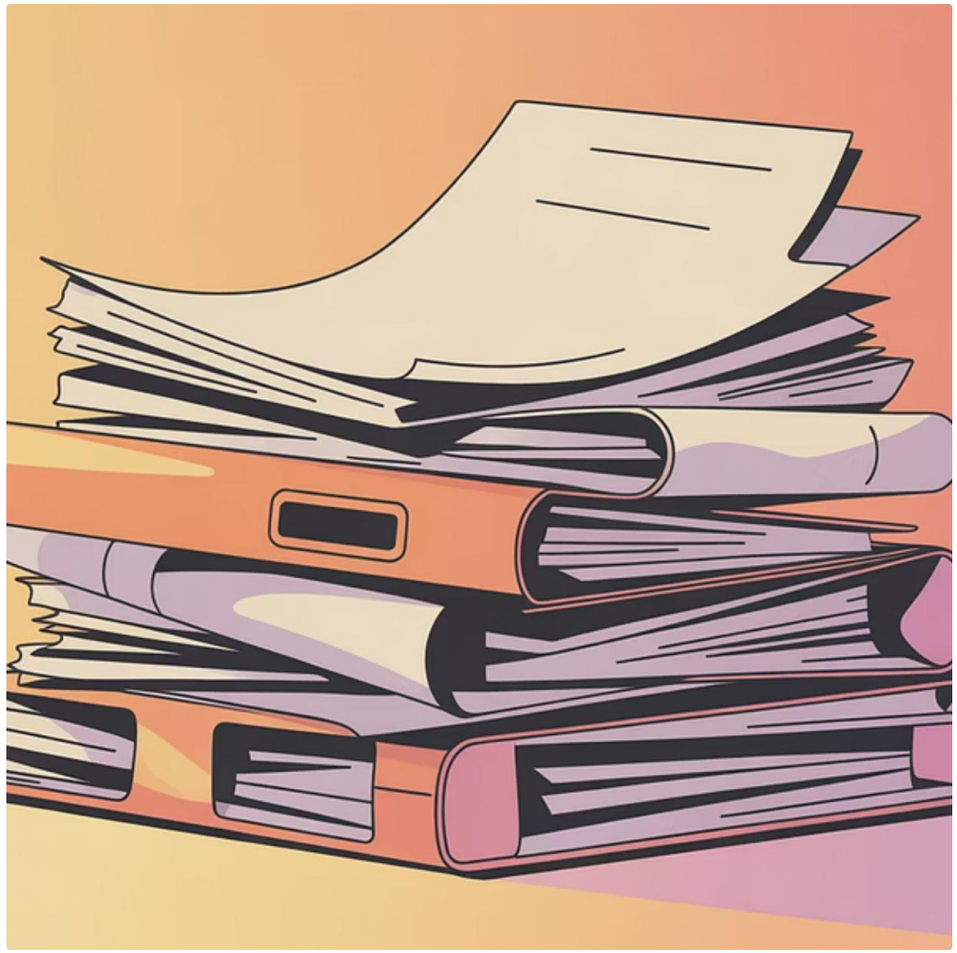
Database Management Systems: Advantages & Architecture

Exploring the power of organized data and its foundational structure.



Why DBMS Over Conventional File Systems?

Unlock superior data management capabilities compared to traditional file-based approaches.



File Systems

Prone to data redundancy and inconsistency, leading to errors and inefficiency.

DBMS

Offers structured data, enhanced security, and efficient access for modern needs.

Key Advantage: Enhanced Data Security

DBMS enforces access controls and encryption to prevent unauthorized access, safeguarding critical information.

Robust Protection

- Granular access permissions
- Data encryption at rest and in transit
- Auditing and logging of data access

Mitigating Risks

Unlike conventional file systems, DBMS scales security measures as user numbers grow, proactively protecting sensitive data from internal and external threats.

Data Integration & Consistency

DBMS centralizes data from multiple sources, providing a unified organizational view and ensuring data integrity.

Unified View

Merges disparate data into a single, cohesive repository, eliminating information silos.

Integrity Constraints

Enforces rules to maintain data accuracy and reduce redundancy, a stark contrast to scattered, error-prone file storage.

Efficient Data Access & Sharing

DBMS supports concurrent multi-user access with optimized queries and indexing, significantly boosting collaboration and productivity.

Concurrent Access

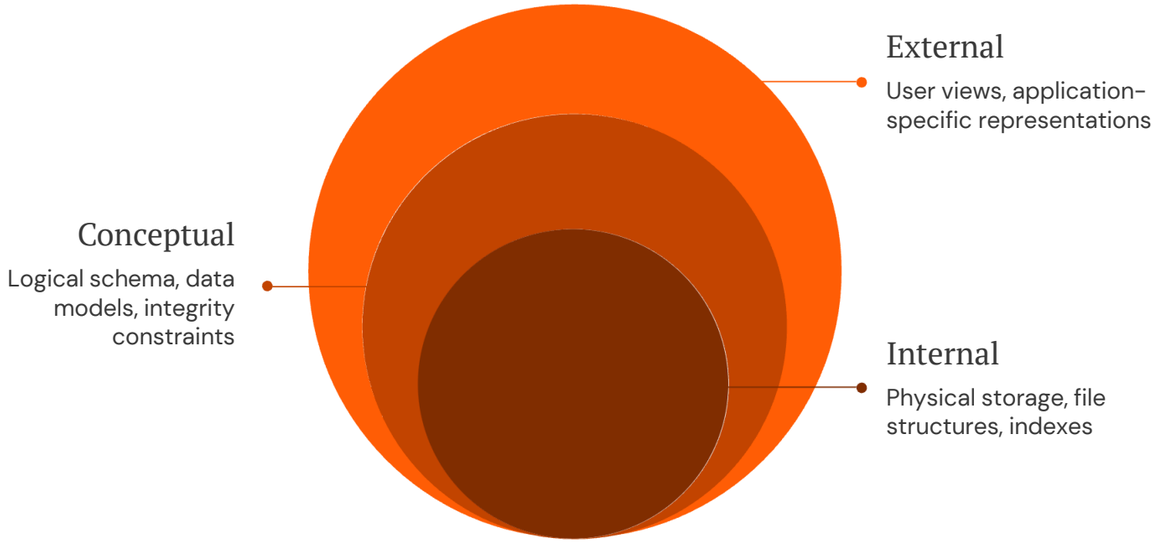
Multiple users can simultaneously access and modify data without conflicts, thanks to sophisticated concurrency control mechanisms.

Seamless Collaboration

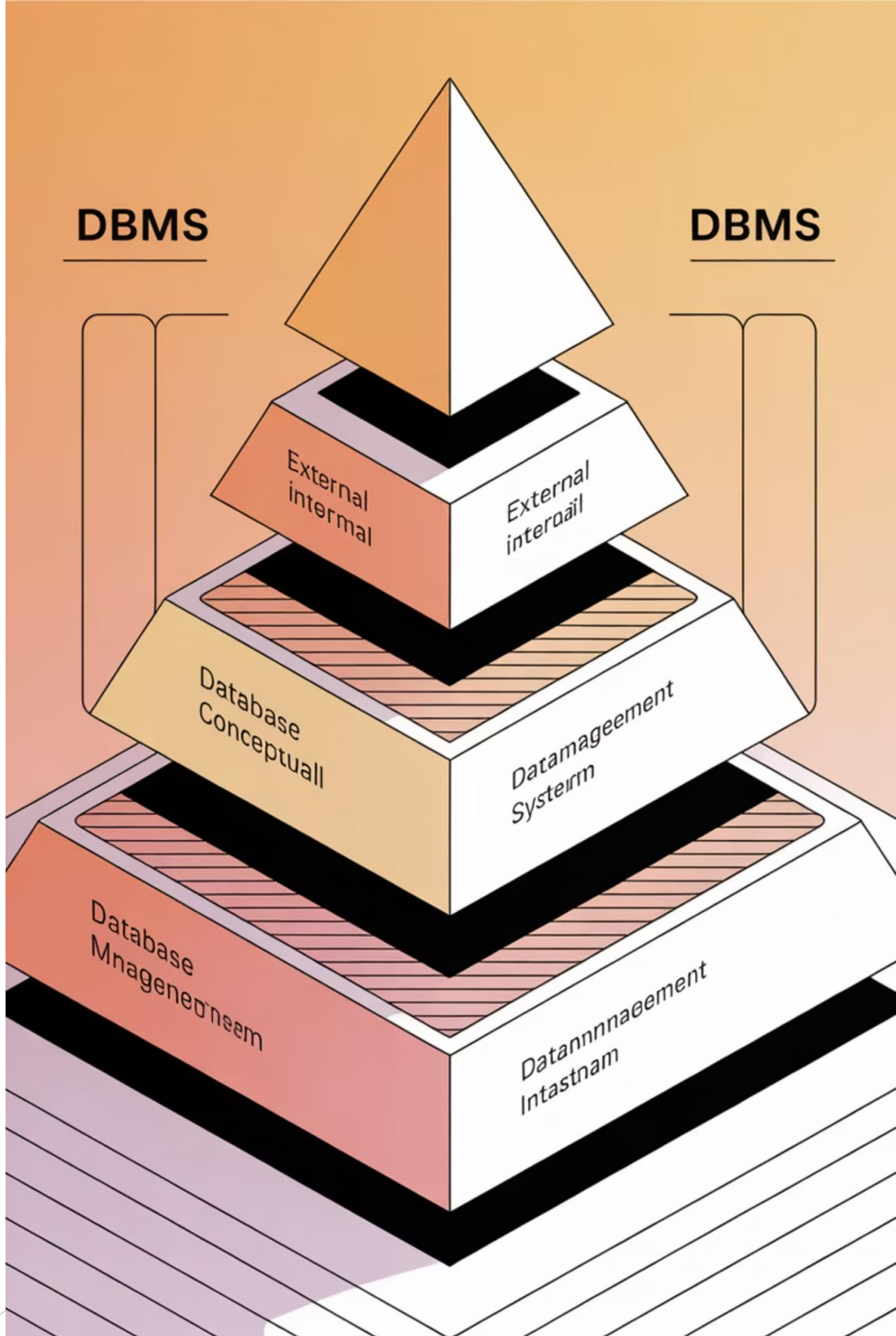
Enables smooth data sharing across various applications and departments, fostering a more collaborative and productive work environment.

Introducing the Three-Level DBMS Architecture

A fundamental framework that separates the user's view from the physical storage of data, ensuring flexibility and independence.



This architectural model is crucial for understanding how modern databases are structured and managed.



The Three Levels Explained

01

External Level (User View)

Provides customized views for different users, displaying only relevant data and hiding complex underlying structures.

02

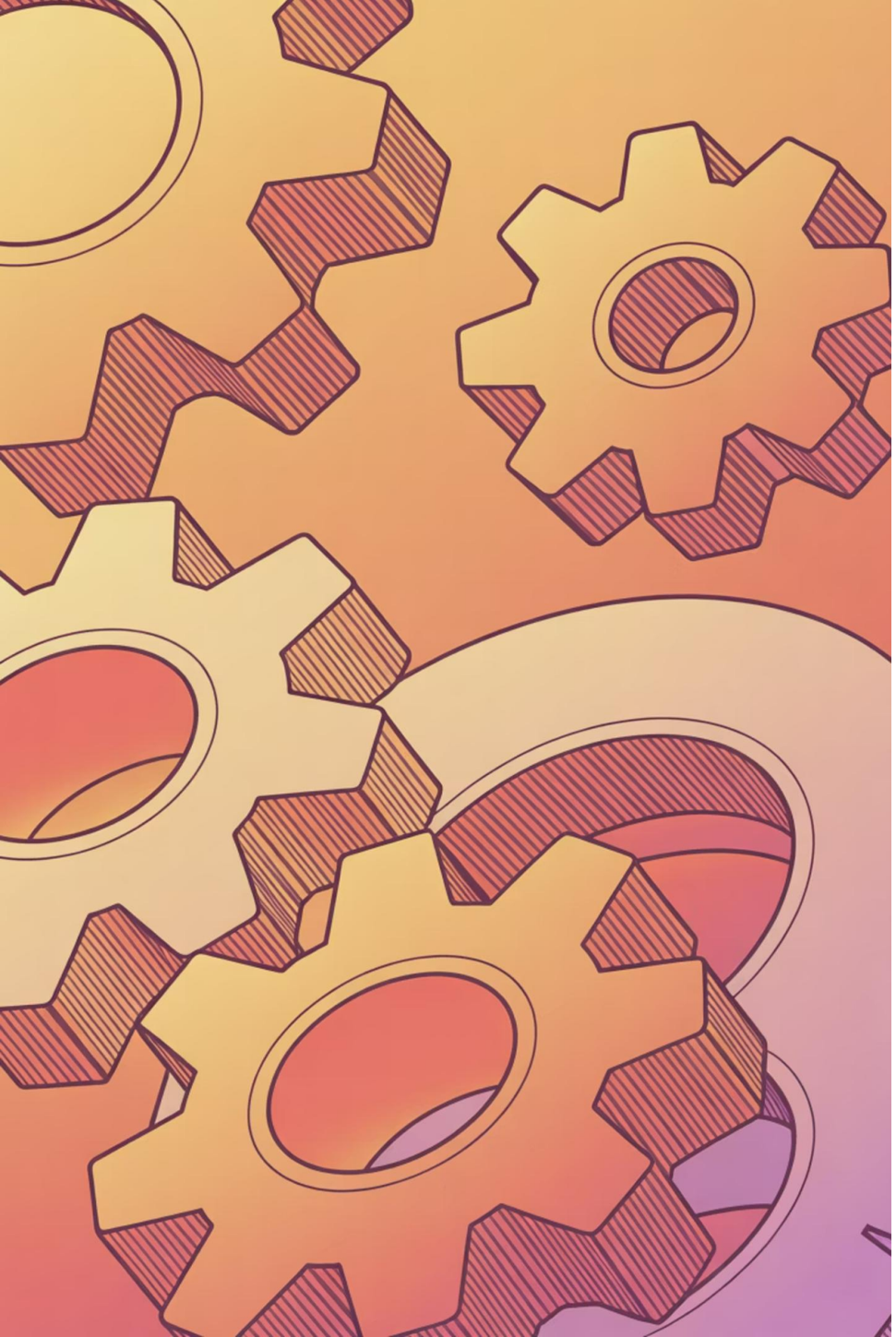
Conceptual Level (Logical View)

Defines what data is stored in the database and the relationships among that data, independent of its physical storage details.

03

Internal Level (Physical View)

Describes how the data is actually stored on hardware, including file structures, indexing, and storage allocation details.



Why This Architecture Matters

This layered approach offers significant benefits, enhancing the database's adaptability and performance.

Data Independence

Separates user interaction from physical data storage, allowing changes in one level without affecting others.

Flexibility & Security

Enables diverse user views and robust security measures, as users only access authorized subsets of data.

Scalability & Performance

Simplifies database management and provides a foundation for enhancing system scalability and overall performance.

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